Deploying Controller Based Wireless Networks Using Fortinet with Lightweight Access Points

Krish Thapaliya

Purpose

This guide/analysis is intended to introduce the Fortinet network system by focusing on how to configure a lightweight access point (FortiAP) with a FortiGate firewall. This guide is created to help the user understand the concepts behind a controller based wireless network and how CAPWAP tunnels work in a real deployment. Setting up a centralized wireless network will give valuable insight into managing SSIDs, isolating wireless traffic, and applying security policies directly from the firewall interface. Gaining these skills is really important for any networking professional who needs to deploy securewireless coverage for both small business and corporate networks.

Background

When setting up a wireless network using Fortinet Firewall, it is important to understand what is an access point architecture. In this type of network architecture, the access point (FortiAP) does not make any major networking decisions on its own. Instead, it will act as a device that relies entirely on a central controller to manage its configuration, push down security policies, and handle user traffic. In a Fortinet based network, the FortiGate firewall has a built in wireless LAN controller inside its operating system. This means that there is no need to have to buy separate, expensive controller hardware to run your wireless network, as the firewall handles everything out of the box.

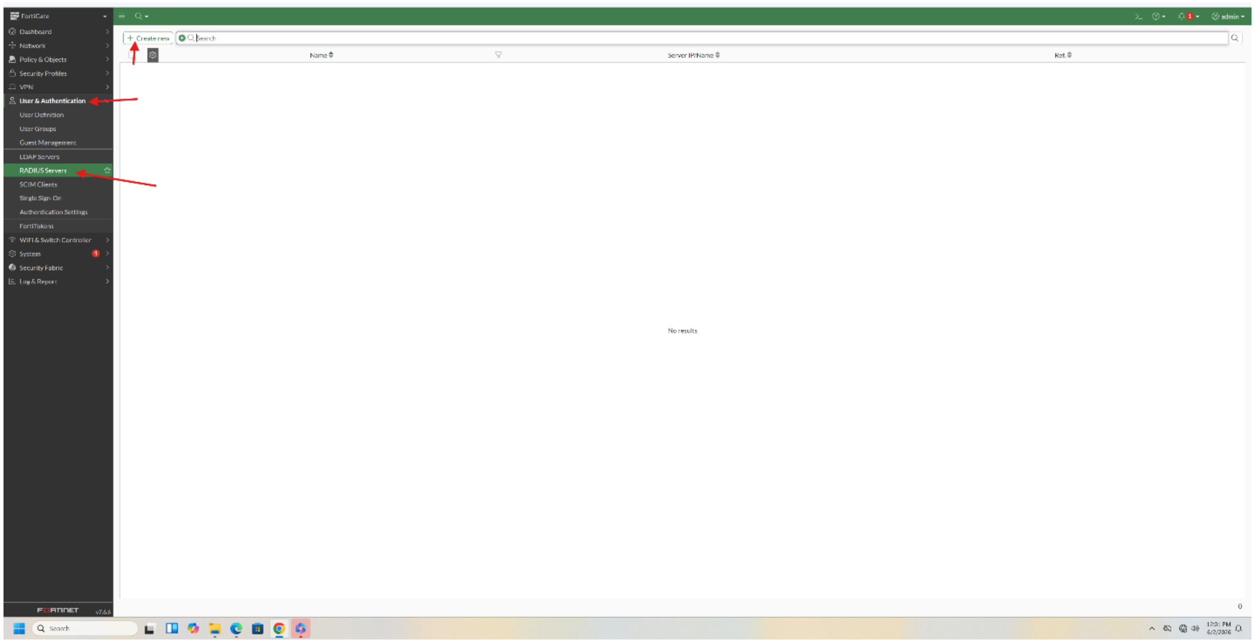
The communication between the FortiGate controller and the lightweight FortiAP relies on a standard industry protocol called CAPWAP (Control and Provisioning of Wireless Access Points). CAPWAP works by establishing two secure tunnels between the firewall and the access point. The first tunnel is the Control Plane, where FortiGate uses to push configurations and monitor the AP's health. The 2nd tunnel is the Data Plane, which takes all the wireless traffic from the users on the Wi Fi and tunnels it directly back to the FortiGate firewall.

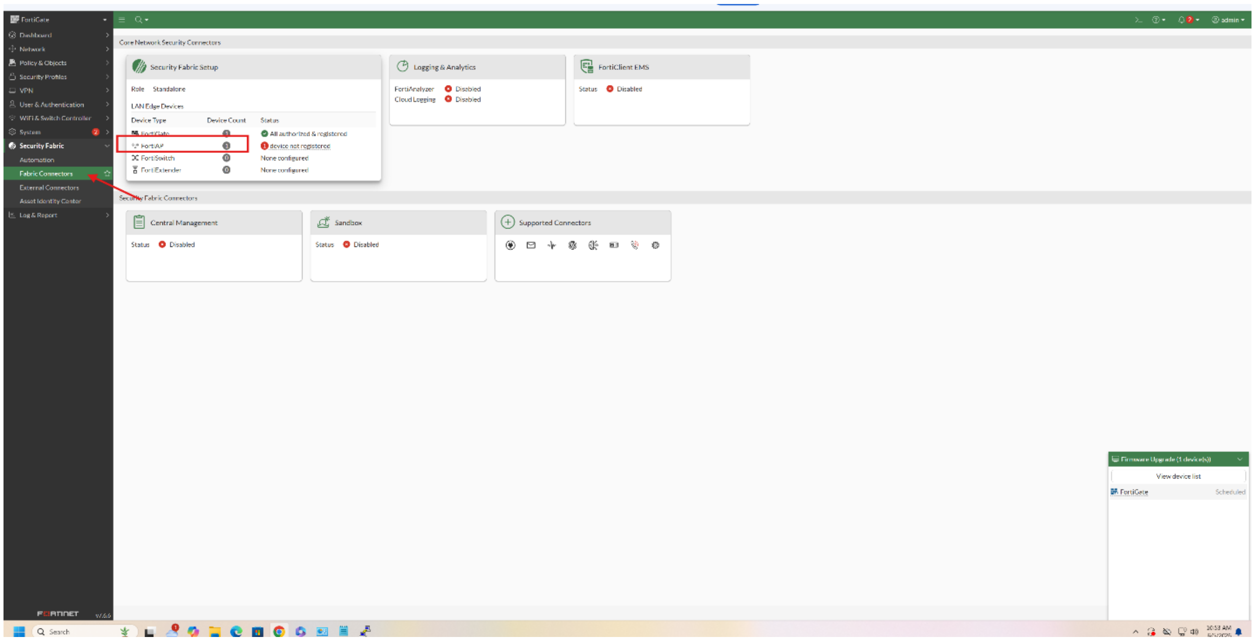
Since all the wireless traffic is tunneled directly to the firewall over the CAPWAP data plane, it created a massive advantage for security and network management. As soon as you create a new wireless network on the FortiGate, it is automatically treated as a local interface on the firewall. This isolates the wireless users from the rest of the internal wired network right away, without you having to manually configure complex Layer 2 VLAN tagging across your office switches. It also allows you to apply the exact same advanced security rules and antivirus tracking to wireless clients that you would normally apply to wired computers.

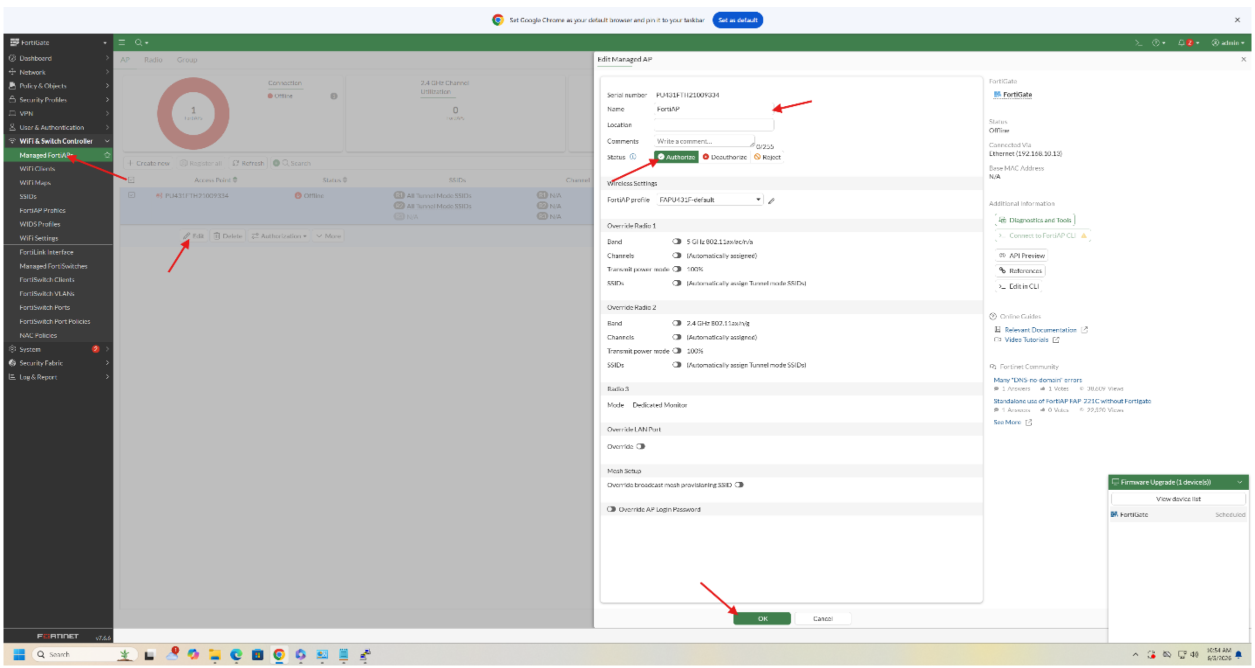
In a normal deployment, like the one in this guide/analysis using a FortiGate 40 F and a FortiAP 421E, the setup process follows a specific lifecycle. First, the lightweight AP is connected to the network and searches for the firewall using methods like DHCP options or broadcast discovery. Once discovered, the FortiAP appears on the FortiGate GUI client as an unmanaged device, where the network administrator has to manually authorize it. After authorization, the FortiGate automatically pushes down the wireless profiles, channel settings, and WPA2 security policies, transforming the standalone hardware into an extension of the firewall's security perimeter.

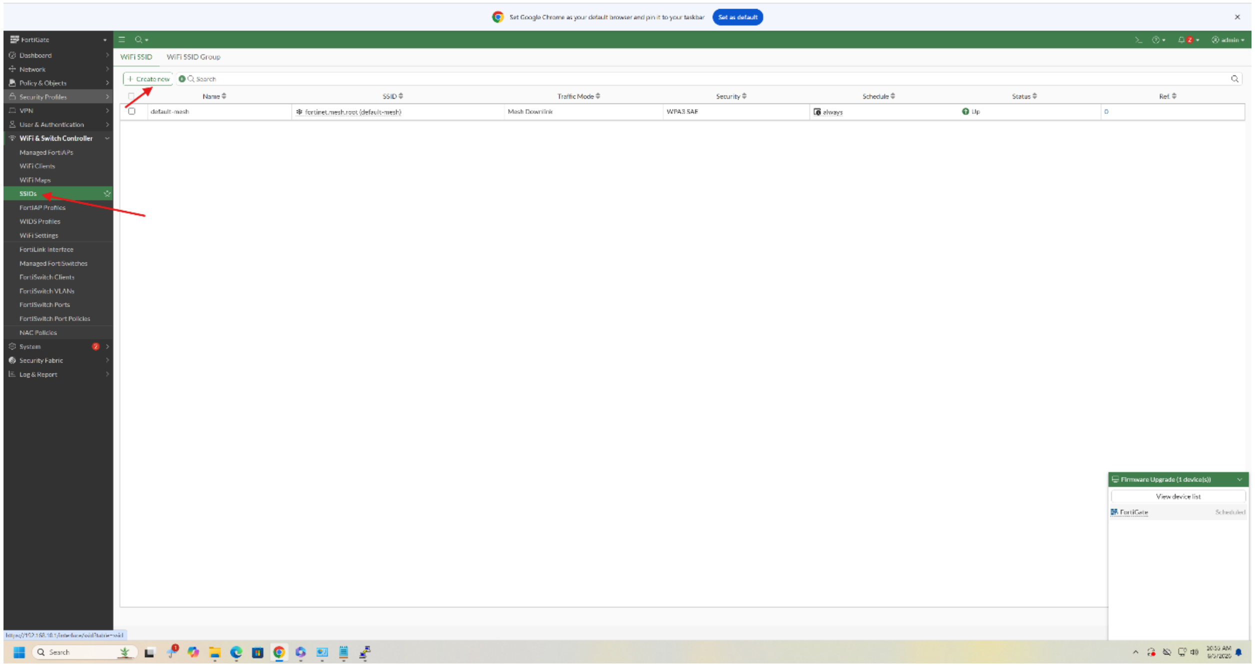
Lab Commands

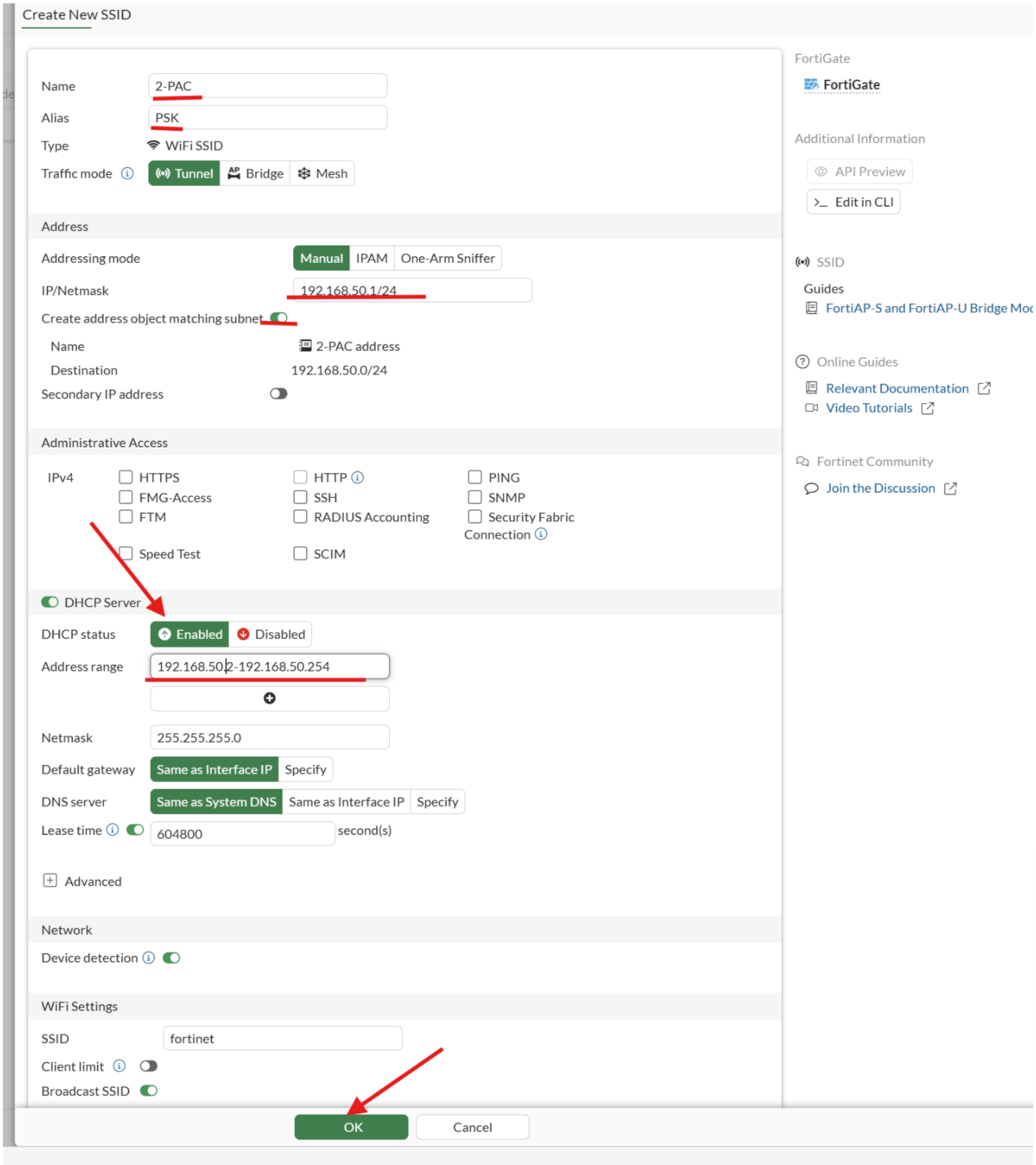
1. Follow the arrows in order as highlighted in each of the images below




Core Network Security Connectors
Security Fabric Setup
Logging & Analytics
FortiClient EMS
Central Management
Sandbox
Supported Connectors


Edit Managed AP
Wireless Settings
Override Radio 1
Override Radio 2
OK
Cancel


Set Google Chrome as your default browser and pin it to your taskbar
Set as default
FortiGate
Dashboard
Network
Policy & Objects
Security Profiles
VPN
User & Authentication
WiFi & Switch Controller
Managed FortiAPs
WiFi Clients
WiFi Maps
SSIDs
FortiAP Profiles
WIDS Profiles
WiFi Settings
FortiLink Interface
Managed FortiSwitches
FortiSwitch Clients
FortiSwitch VLANs
FortiSwitch Ports
FortiSwitch Port Policies
NAC Policies
System
Security Fabric
Log & Report
WiFi SSID
WiFi SSID Group
Create new
Search
Name
SSID
Traffic Mode
Security
Schedule
Status
Ref.
default-mesh
fortinet.mesh.root (default-mesh)
Mesh Downlink
WPA3 SAE
always
Up
0
Firmware Upgrade (1 device(s))
View device list
FortiGate
Scheduled



Create New SSID
Name
2-PAC
Alias
PSK
Type
WiFi SSID
Traffic mode
Tunnel
Bridge
Mesh
Address
Addressing mode
Manual
IPAM
One-Arm Sniffer
IP/Netmask
192.168.50.1/24
Create address object matching subnet
Name
2-PAC address
Destination
192.168.50.0/24
Secondary IP address
Administrative Access
IPv4
HTTPS
HTTP
PING
FMG-Access
SSH
SNMP
FTM
RADIUS Accounting
Security Fabric Connection
Speed Test
SCIM
DHCP Server
DHCP status
Enabled
Disabled
Address range
192.168.50.2-192.168.50.254
Netmask
255.255.255.0
Default gateway
Same as Interface IP
Specify
DNS server
Same as System DNS
Same as Interface IP
Specify
Lease time
604800
second(s)
Advanced
Network
Device detection
WiFi Settings
SSID
fortinet
Client limit
Broadcast SSID
OK
Cancel
FortiGate
FortiGate
Additional Information
API Preview
Edit in CLI
SSID
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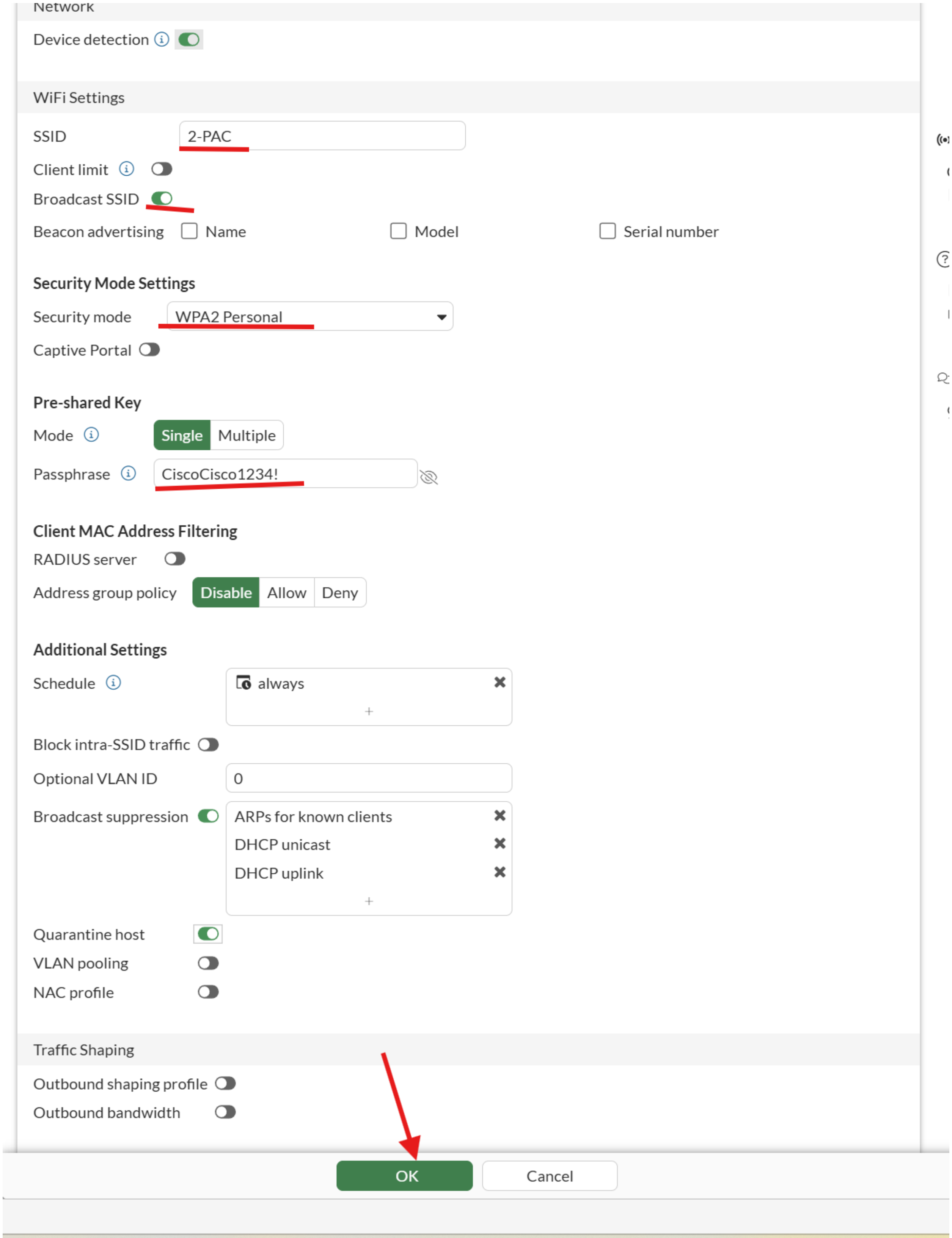
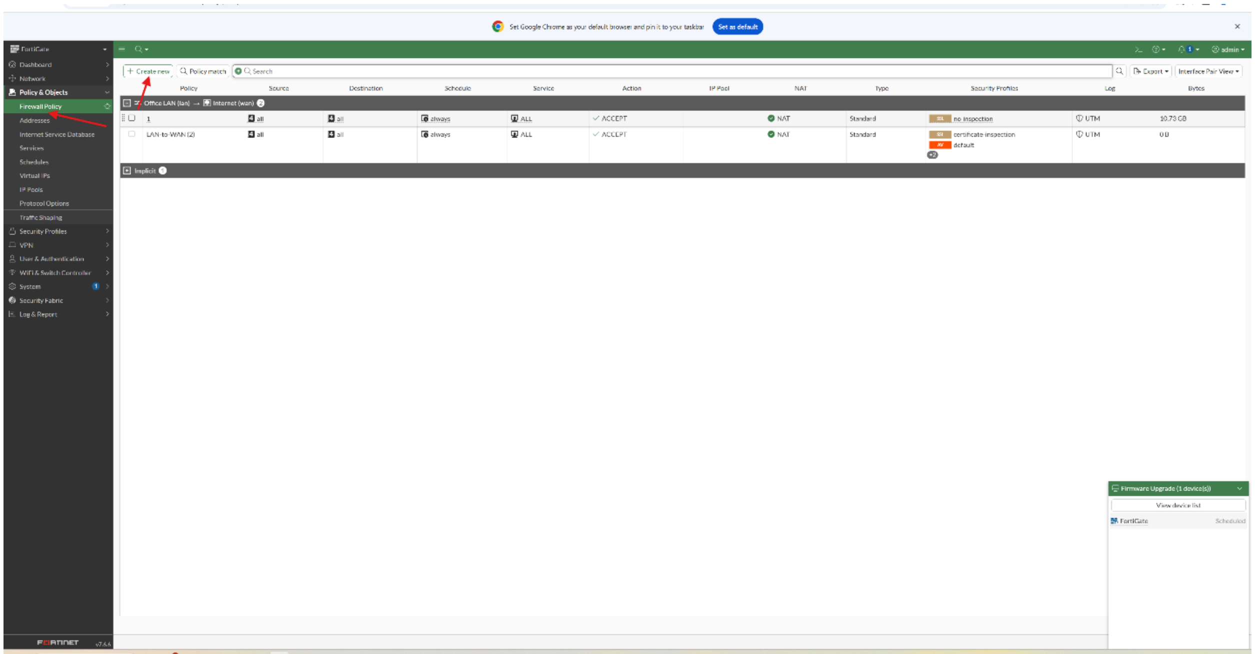
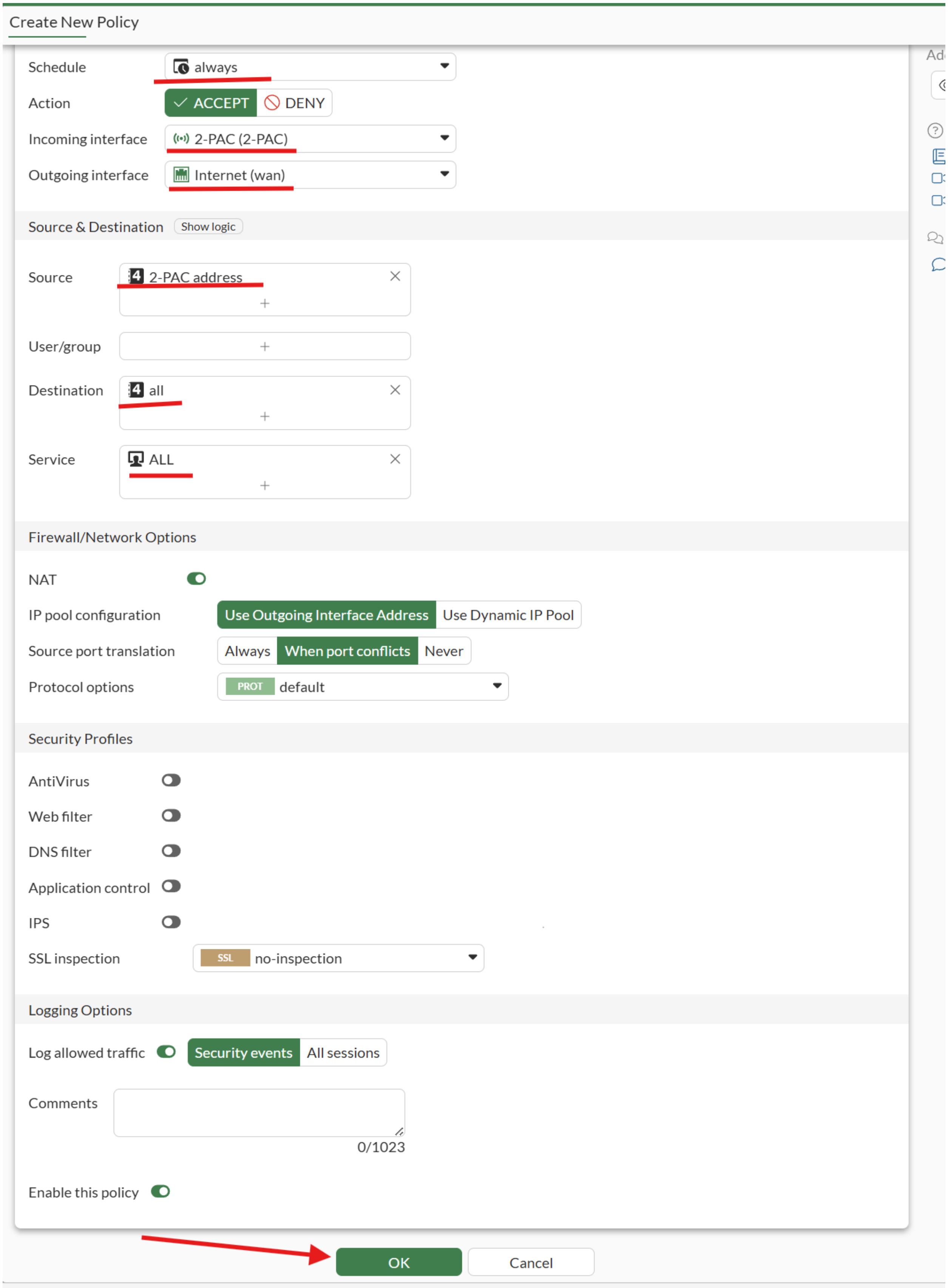
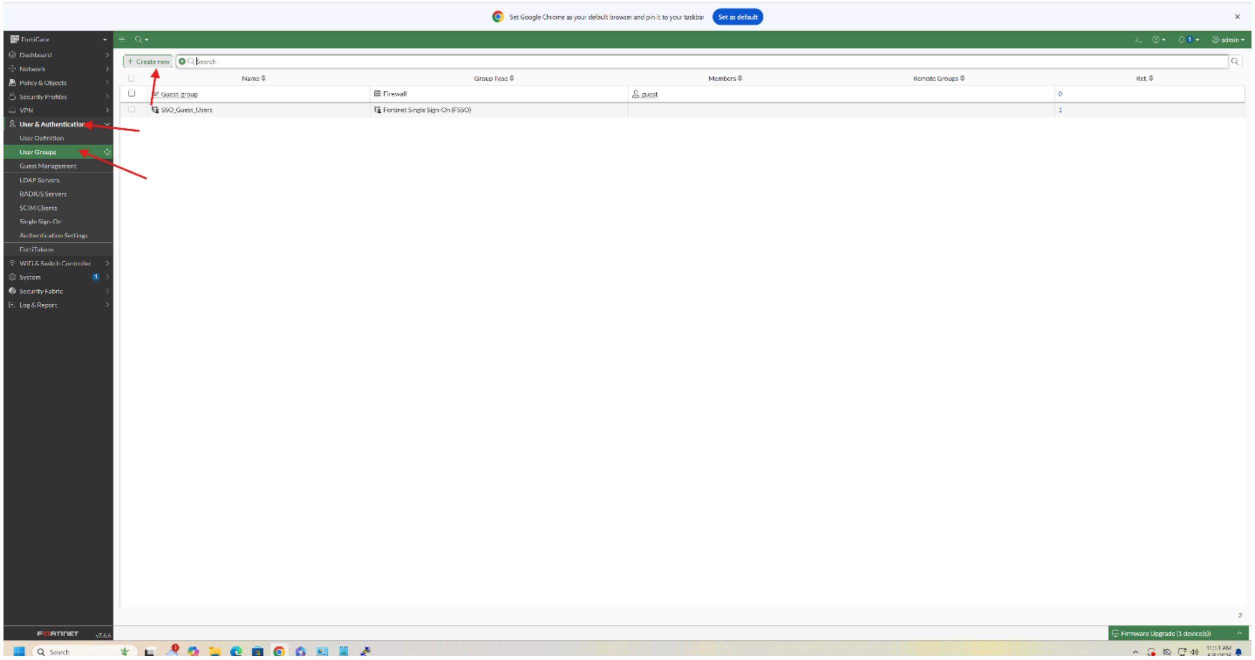


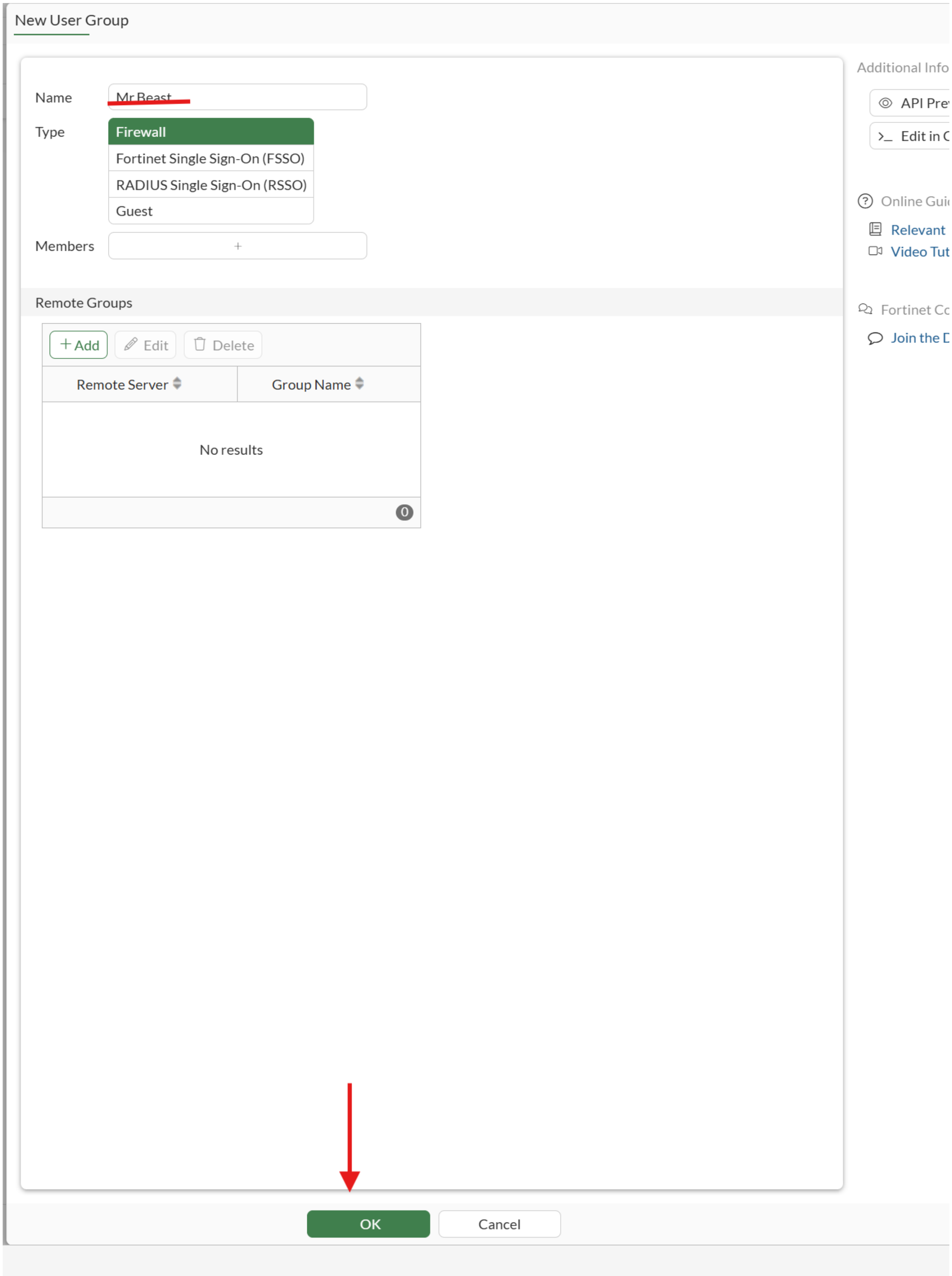
Network
Device detection
WiFi Settings
SSID
2-PAC
Client limit
Broadcast SSID
Beacon advertising
Name
Model
Serial number
Security Mode Settings
Security mode
WPA2 Personal
Captive Portal
Pre-shared Key
Mode
Single
Multiple
Passphrase
CiscoCisco1234!
Client MAC Address Filtering
RADIUS server
Address group policy
Disable
Allow
Deny
Additional Settings
Schedule
always
Block intra-SSID traffic
Optional VLAN ID
0
Broadcast suppression
ARPs for known clients
DHCP unicast
DHCP uplink
Quarantine host
VLAN pooling
NAC profile
Traffic Shaping
Outbound shaping profile
Outbound bandwidth
OK
Cancel


Set Google Chrome as your default browser and pin it to your taskbar
Set as default
FortiGate
Dashboard
Network
Policy & Objects
Firewall Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options
Traffic Shaping
Security Profiles
VPN
User & Authentication
WiFi & Switch Controller
System
Security Fabric
Log & Report
+ Create new
Policy match
Search
Export
Interface Pair View
Policy
Source
Destination
Schedule
Service
Action
IP Pool
NAT
Type
Security Profiles
Log
Bytes
Office LAN (lan) → Internet (wan)
1
all
all
always
ALL
ACCEPT
NAT
Standard
no-inspection
UTM
10.73 GB
LAN-to-WAN (2)
certificate-inspection
default
0 B
Implicit
Firmware Upgrade (1 device(s))
View device list
FortiGate
Scheduled
FORTINET



Create New Policy
Schedule
always
Action
ACCEPT
DENY
Incoming interface
2-PAC (2-PAC)
Outgoing interface
Internet (wan)
Source & Destination
Show logic
Source
2-PAC address
User/group
Destination
all
Service
ALL
Firewall/Network Options
NAT
IP pool configuration
Use Outgoing Interface Address
Use Dynamic IP Pool
Source port translation
Always
When port conflicts
Never
Protocol options
PROT
default
Security Profiles
AntiVirus
Web filter
DNS filter
Application control
IPS
SSL inspection
SSL
no-inspection
Logging Options
Log allowed traffic
Security events
All sessions
Comments
0/1023
Enable this policy
OK
Cancel




New User Group
Name
Mr Beast
Type
Firewall
Fortinet Single Sign-On (FSSO)
RADIUS Single Sign-On (RSSO)
Guest
Members
Remote Groups
+ Add
Edit
Delete
Remote Server
Group Name
No results
0
OK
Cancel
Additional Info
API Pre
Edit in C
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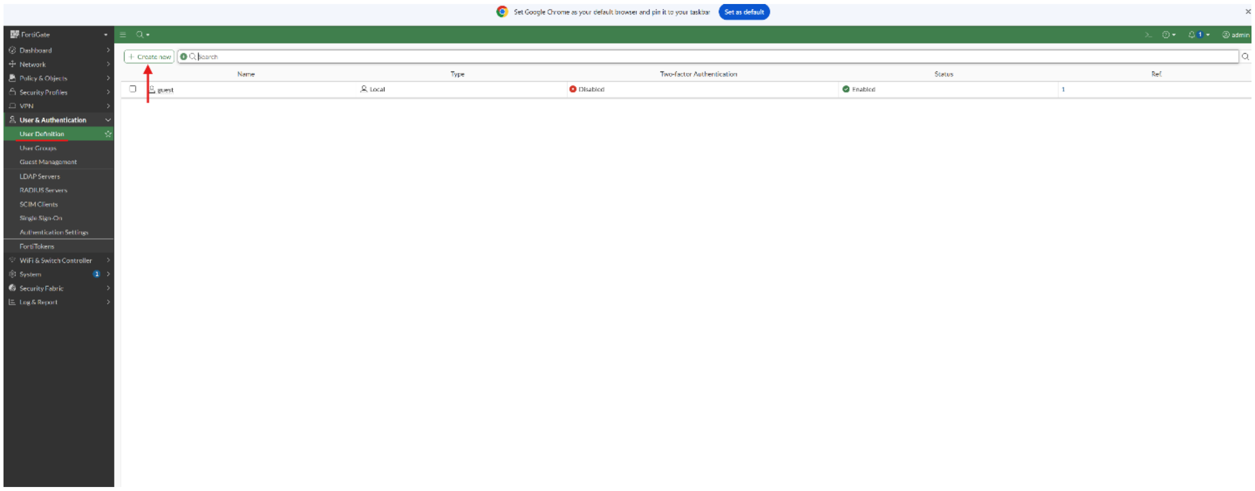
Relevant
Video Tut
Fortinet Cc
Join the D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Set as default
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Network
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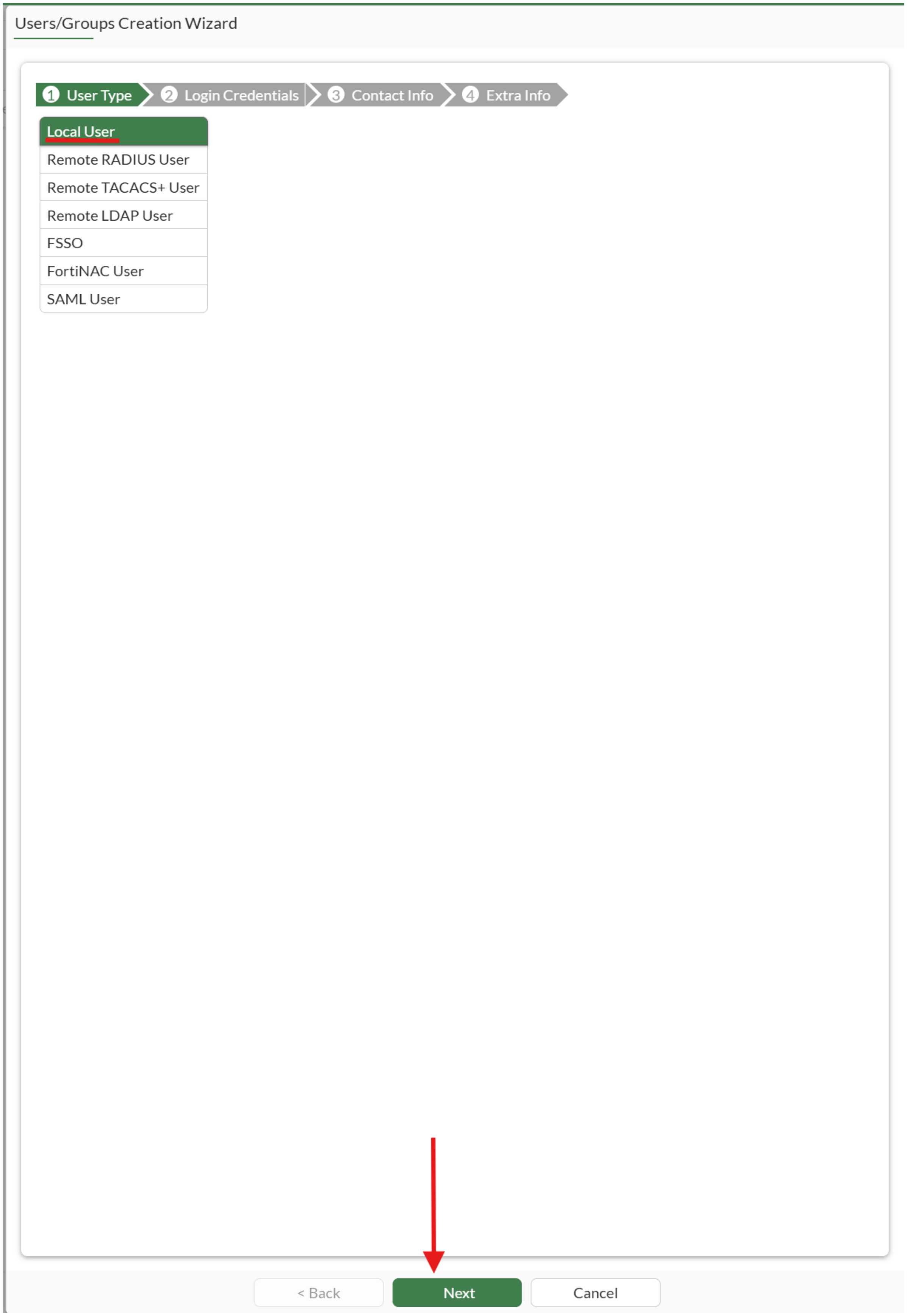
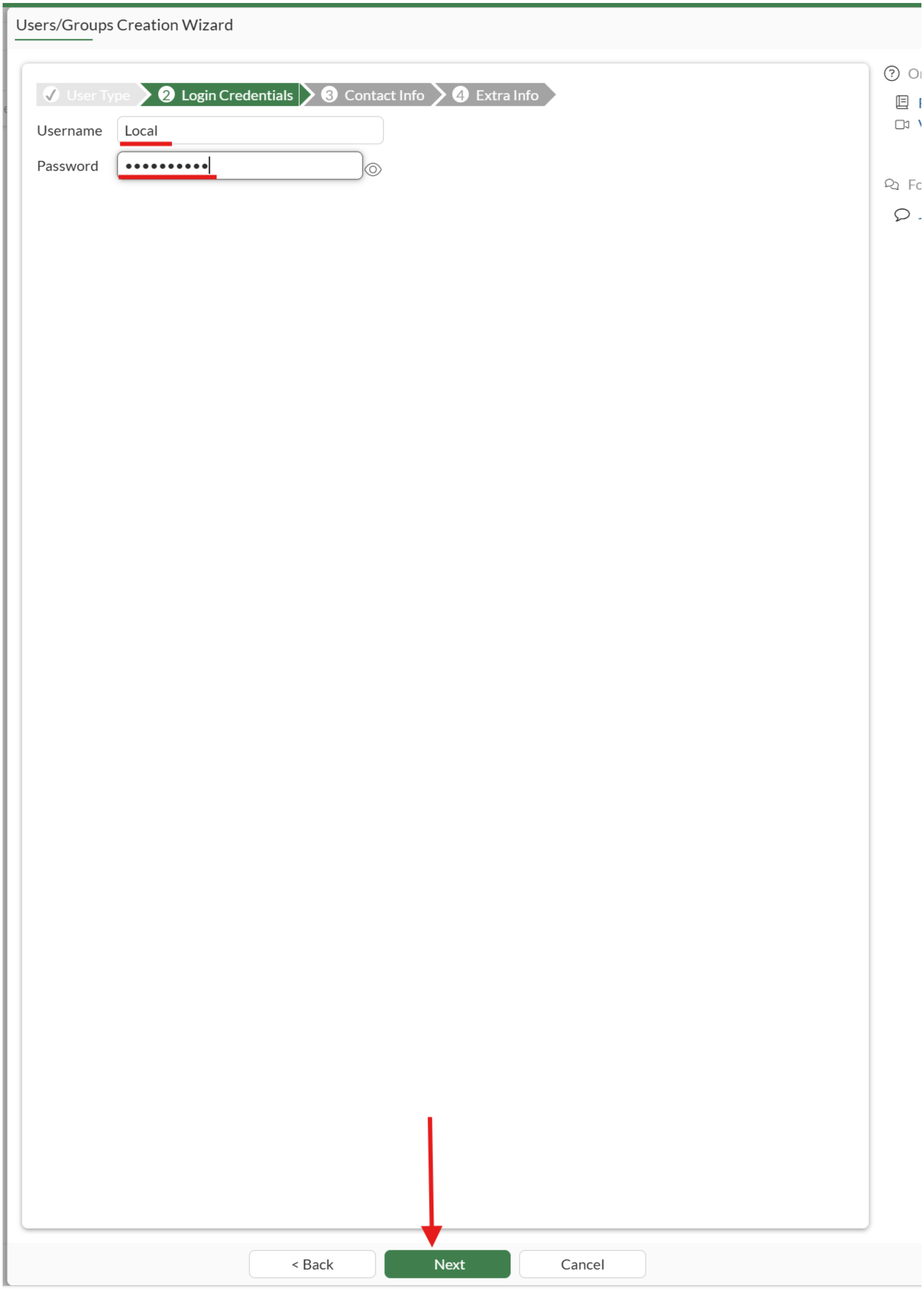
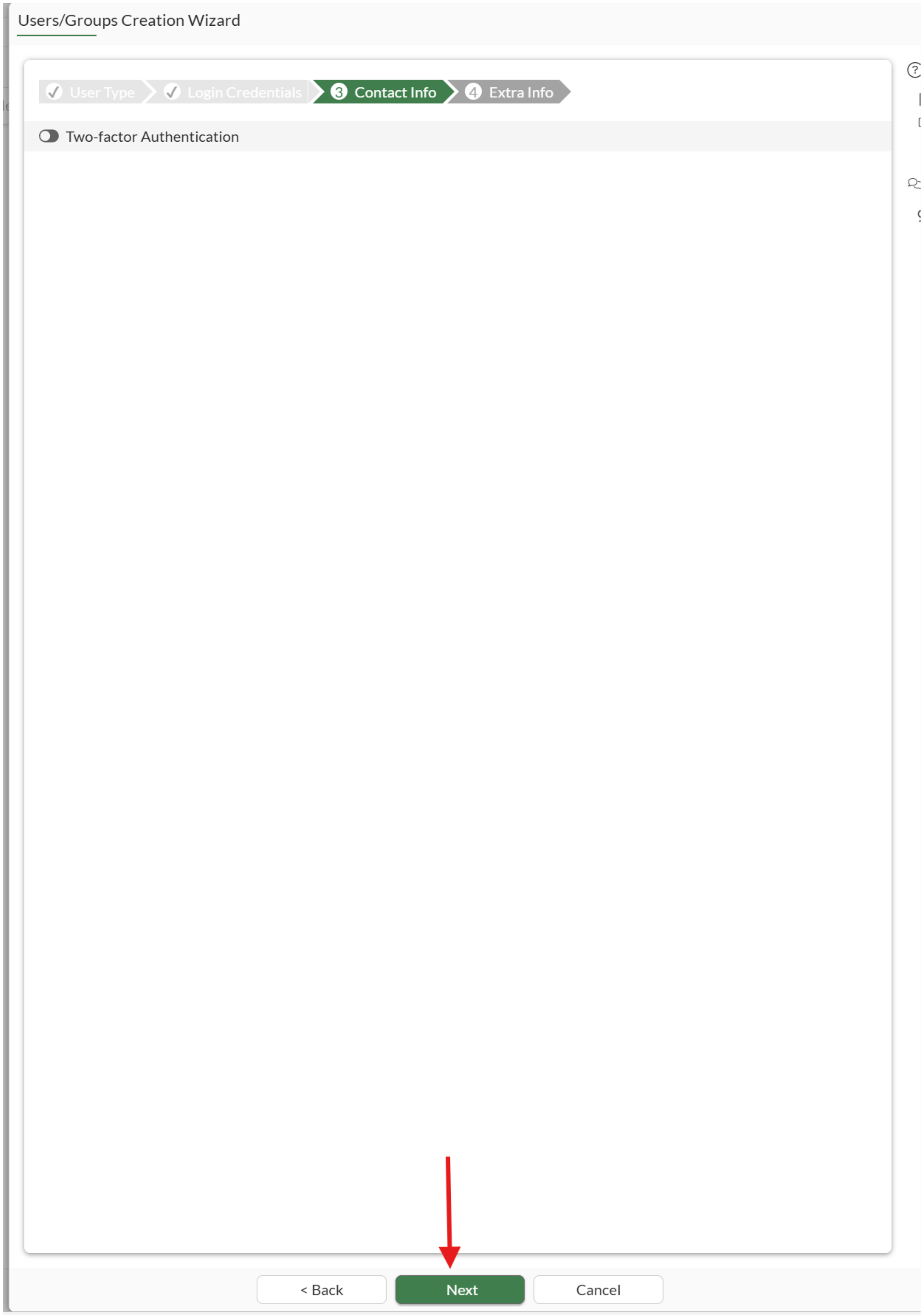
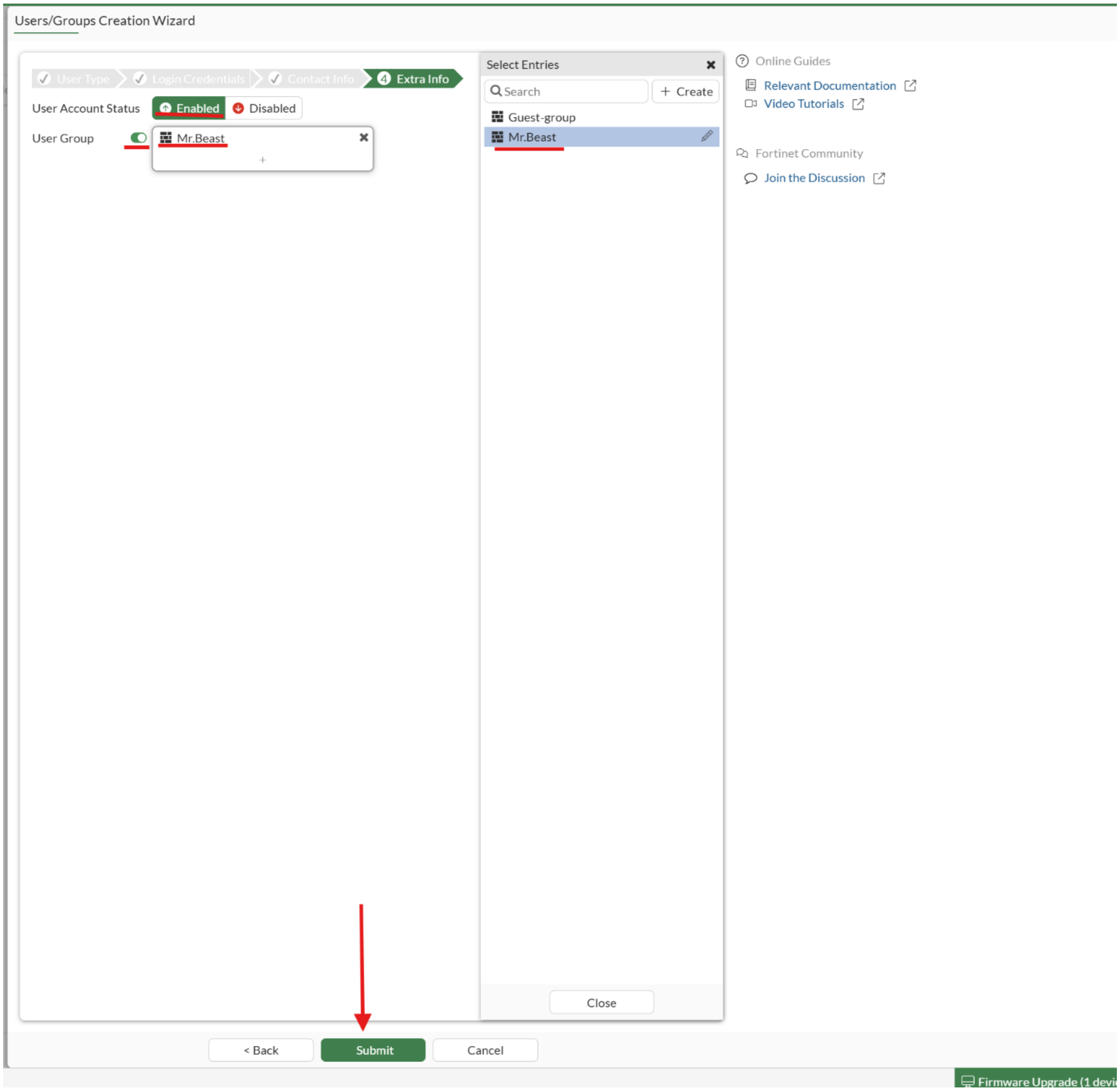
VPN
User & Authentication
User Definition
User Groups
Guest Management
LDAP Servers
RADIUS Servers
SCIM Clients
Single Sign-On
Authentication Settings
FortiTokens
WiFi & Switch Controller
System
Security Fabric
Log & Report
admin
Create New
Search
Name
Type
Two-factor Authentication
Status
Ref
guest
Local
Disabled
Enabled
1


Users/Groups Creation Wizard
1 User Type
2 Login Credentials
3 Contact Info
4 Extra Info
Local User
Remote RADIUS User
Remote TACACS+ User
Remote LDAP User
FSSO
FortiNAC User
SAML User
< Back
Next
Cancel


Users/Groups Creation Wizard
User Type
2 Login Credentials
3 Contact Info
4 Extra Info
Username
Local
Password
< Back
Next
Cancel


Users/Groups Creation Wizard
User Type
Login Credentials
3 Contact Info
4 Extra Info
Two-factor Authentication
< Back
Next
Cancel


Users/Groups Creation Wizard
User Type
Login Credentials
Contact Info
4 Extra Info
User Account Status
Enabled
Disabled
User Group
Mr.Beast
Select Entries
Search
+ Create
Guest-group
Mr.Beast
Close
< Back
Submit
Cancel
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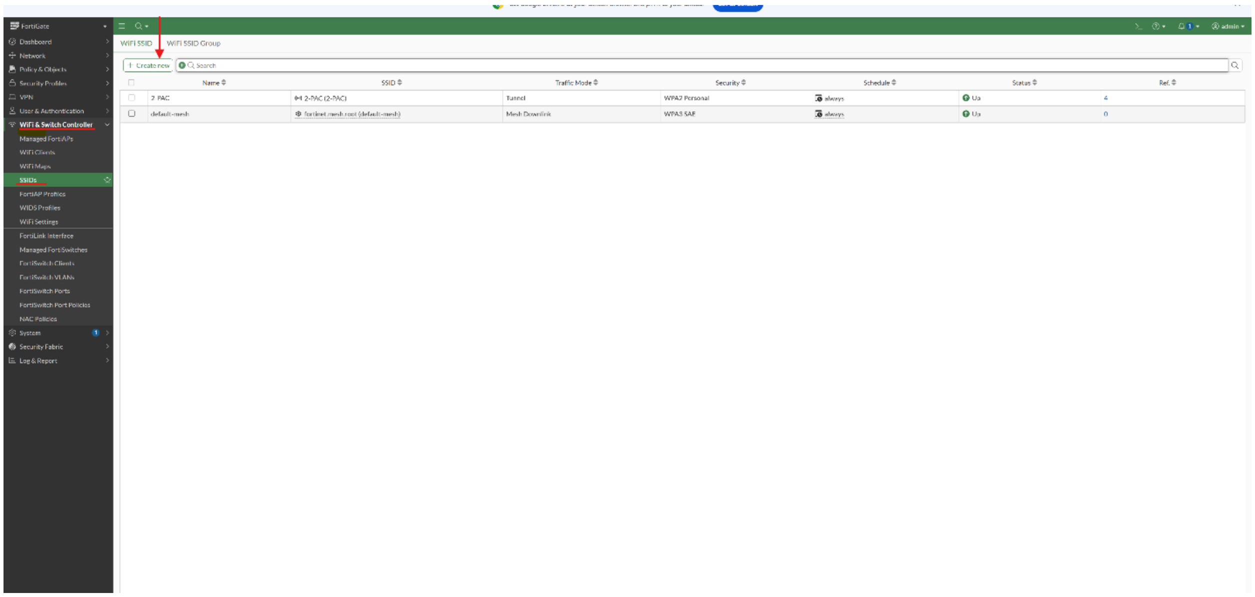

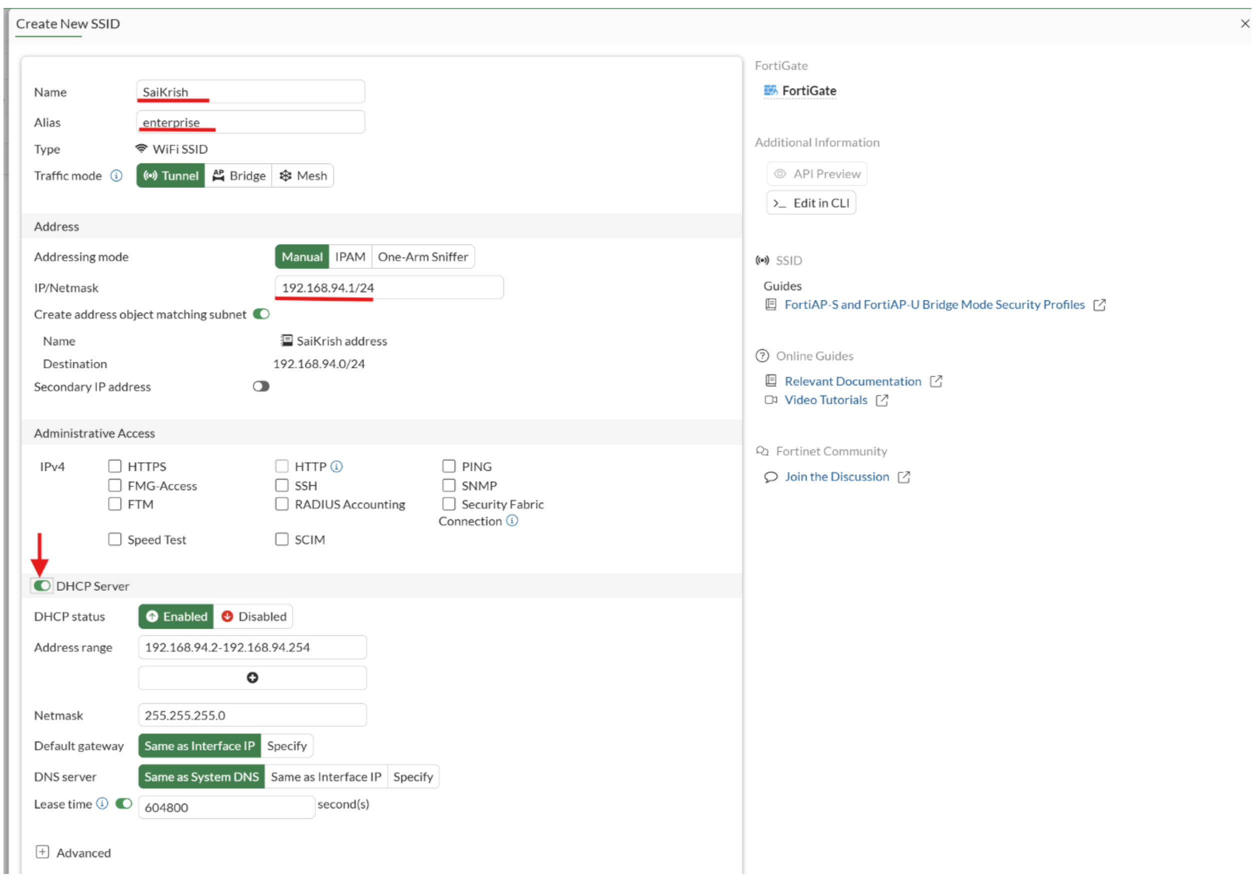
Create New SSID
Name
SaiKrish
Alias
enterprise
Type
WiFi SSID
Traffic mode
Tunnel
Bridge
Mesh
Address
Addressing mode
Manual
IPAM
One-Arm Sniffer
IP/Netmask
192.168.94.1/24
Create address object matching subnet
Name
SaiKrish address
Destination
192.168.94.0/24
Secondary IP address
Administrative Access
IPv4
HTTPS
HTTP
PING
FMG-Access
SSH
SNMP
FTM
RADIUS Accounting
Security Fabric Connection
Speed Test
SCIM
DHCP Server
DHCP status
Enabled
Disabled
Address range
192.168.94.2-192.168.94.254
Netmask
255.255.255.0
Default gateway
Same as Interface IP
Specify
DNS server
Same as System DNS
Same as Interface IP
Specify
Lease time
604800
second(s)
Advanced
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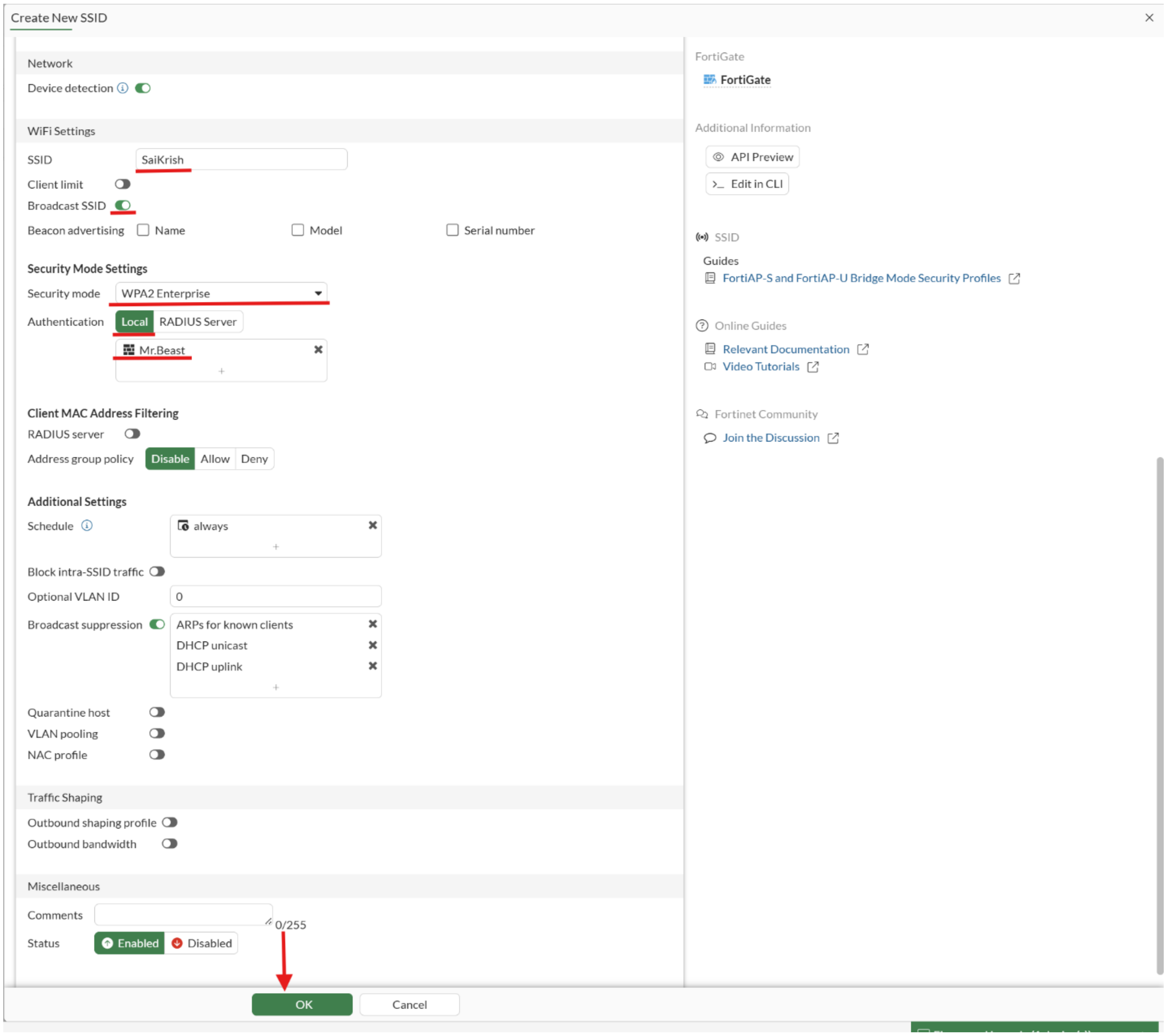
Create New SSID
Network
Device detection
WiFi Settings
SSID
SaiKrish
Client limit
Broadcast SSID
Beacon advertising
Name
Model
Serial number
Security Mode Settings
Security mode
WPA2 Enterprise
Authentication
Local
RADIUS Server
Mr.Beast
Client MAC Address Filtering
RADIUS server
Address group policy
Disable
Allow
Deny
Additional Settings
Schedule
always
Block intra-SSID traffic
Optional VLAN ID
0
Broadcast suppression
ARPs for known clients
DHCP unicast
DHCP uplink
Quarantine host
VLAN pooling
NAC profile
Traffic Shaping
Outbound shaping profile
Outbound bandwidth
Miscellaneous
Comments
0/255
Status
Enabled
Disabled
OK
Cancel
FortiGate
FortiGate
Additional Information
API Preview
Edit in CLI
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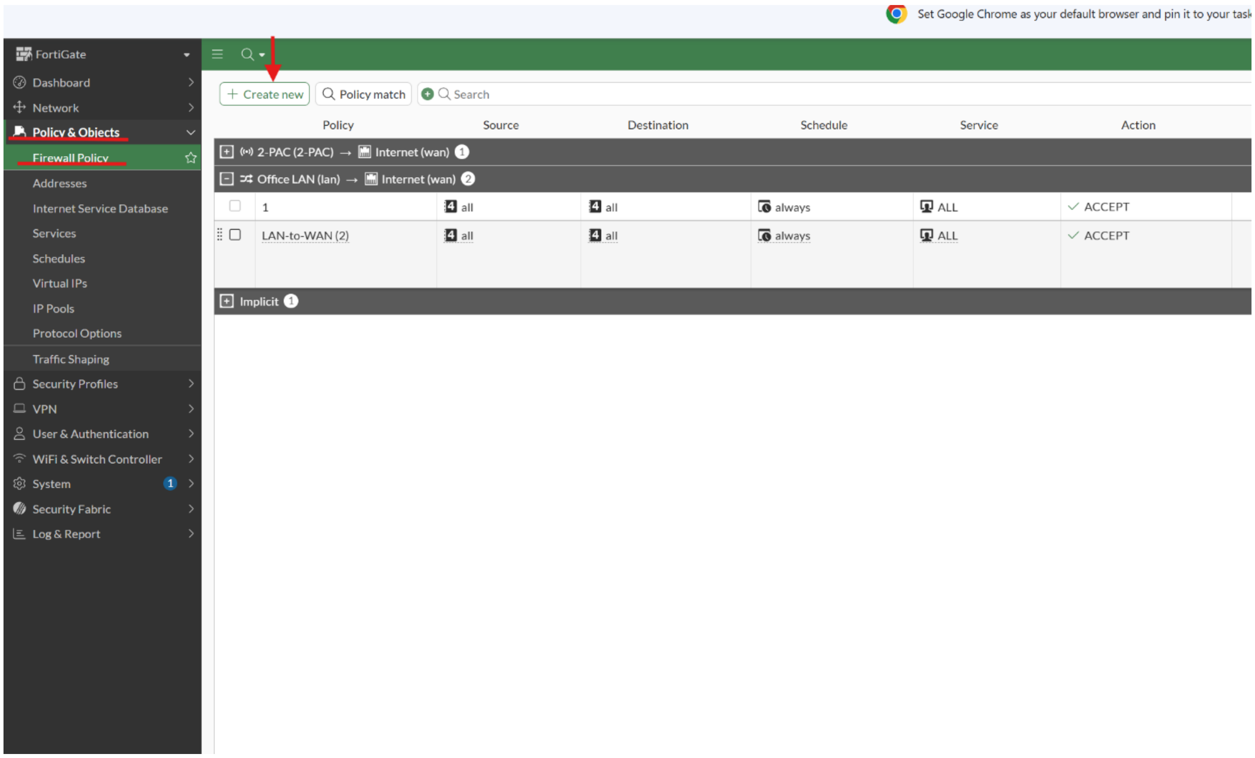

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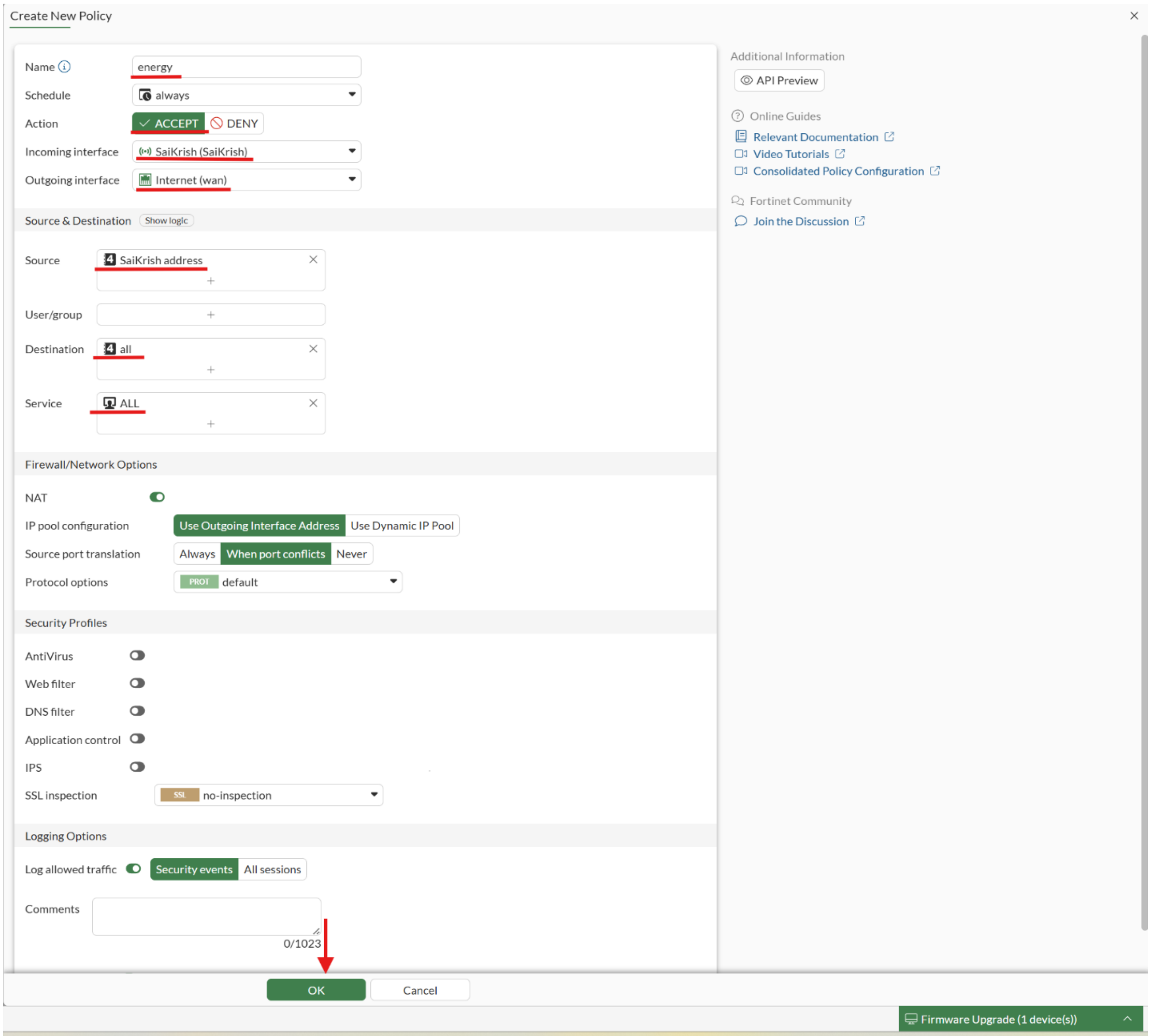
Security Profiles
VPN
User & Authentication
WiFi & Switch Controller
System
Security Fabric
Log & Report
Create new
Policy match
Search
Policy
Source
Destination
Schedule
Service
Action
2-PAC (2-PAC) → Internet (wan) 1
Office LAN (lan) → Internet (wan) 2
1
all
all
always
ALL
ACCEPT
LAN-to-WAN (2)
all
all
always
ALL
ACCEPT
Implicit 1



Problems

During the guide/analysis setup, I ran into a small physical problem. I found out that the FortiAP cannot be plugged directly into the FortiGate without an external power source, because the FortiGate interface does not provide power over ethernet (PoE). I fixed this issue by adding a standard PoE injector inline between the firewall port and the access point.

Conclusion

In conclusion, this guide/analysis provided a great experience into the Fortinet wireless ecosystem and how a FortiGate firewall functions as a centralized LAN controller. I now have a solid understanding of how lightweight access points use CAPWAP tunnels to pass user traffic back to a security gateway and how to deploy WPA2 security profiles across physical spaces.